

ASSIGNMENT 01

Site Modeling with LiDAR and Physical Fabrication

SITE SELECTION AND SCANNING

Gather LiDAR Data:

Websites:

- National: <https://portal.opentopography.org/datasets>
- Montana: <https://experience.arcgis.com/experience/59eb1cfca29445a0ae04655379c314e9/page/View%2C-Download%2C-and-Request-Data>
- **Washington: <https://lidarportal.dnr.wa.gov/#47.62437:-122.51010:16>**

SITE: BAINBRIDGE ISLAND FERRY TERMINAL

Coordinates
0000, 00000



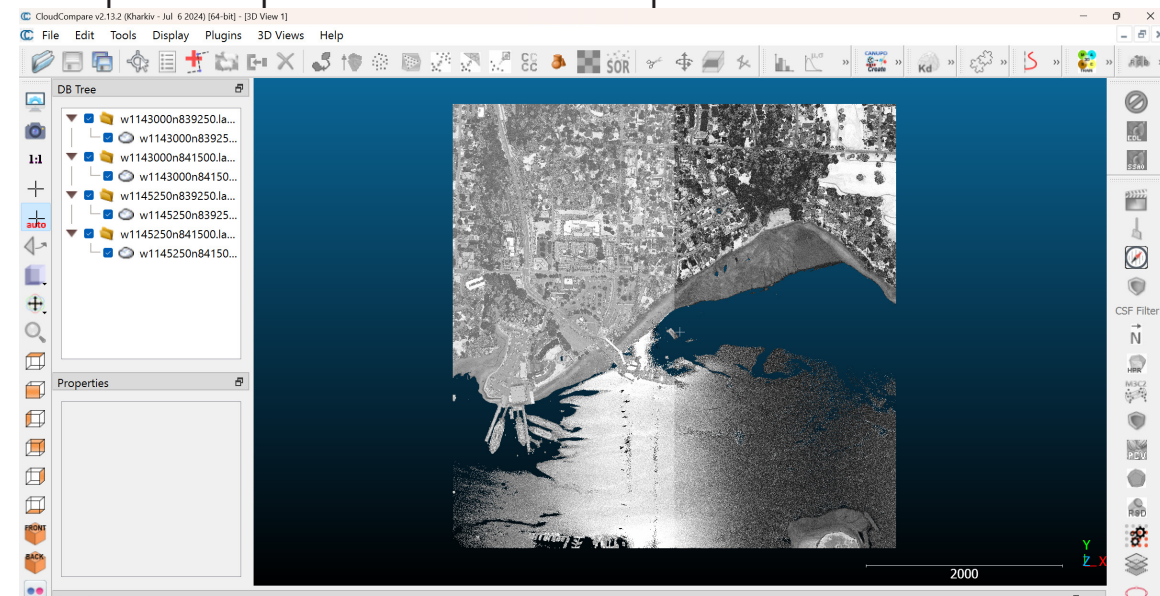
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DATA PROCESSING AND DIGITAL MODELING

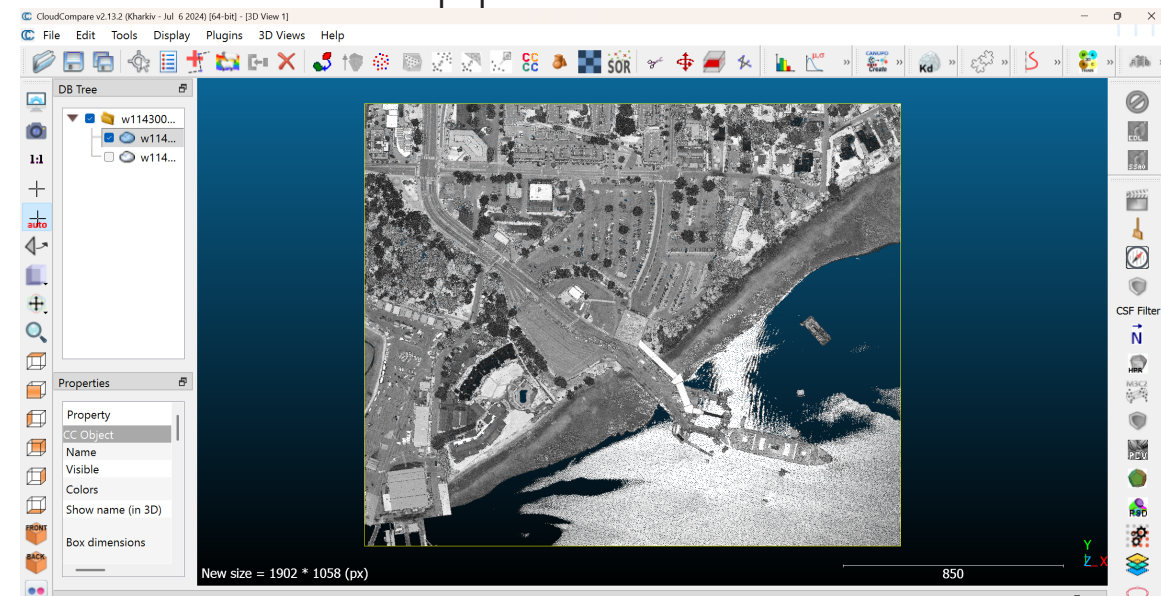
Cloud Compare Process:

1. Import 4 quadrants of LiDAR point cloud data



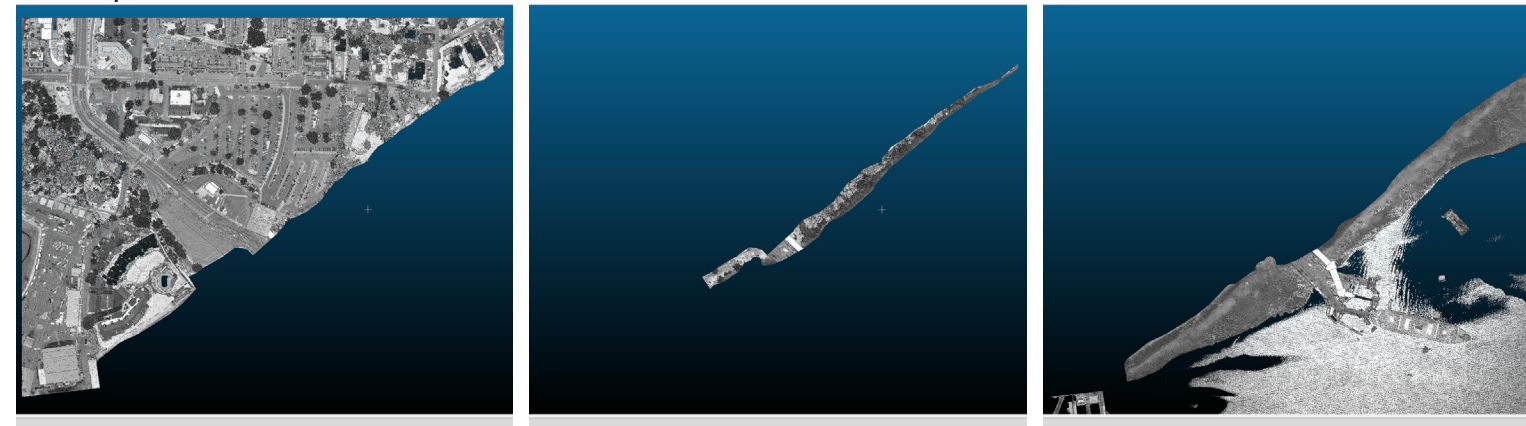
The LiDAR data available to me was from 2024 but downloaded as 4 quadrants in order to create the site I need.

2. Combine and crop point clouds



I combined using the merge tool and the segment tool to combine then crop the point cloud down to the site I needed.

3. Separate sections of data



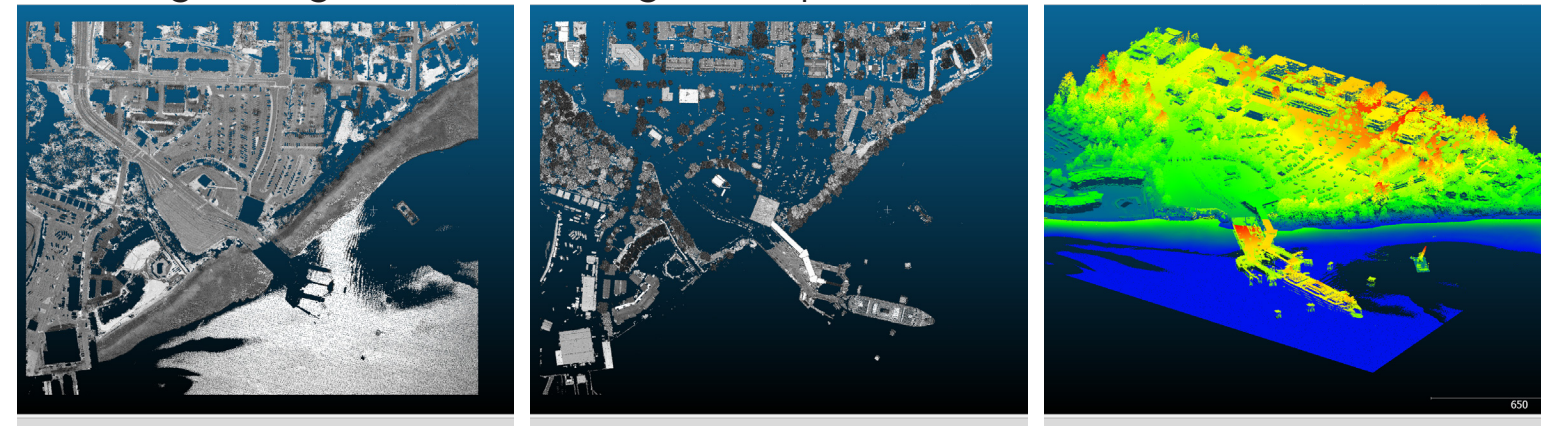
Soft Sloped Land

Steep Bank

Flat Beach+Water

To categorize the ground points vs the off-ground points as best as possible, I segmented the site based on the slope before running the CSF filter independently for each segment.

4. Categorize ground and off ground points



Ground Points

Off Ground Points

The steep bank down to the beach was hardest to separate due to the density of the trees. I also went back in a segmented the dock out because the CSF filter categorized it as ground.

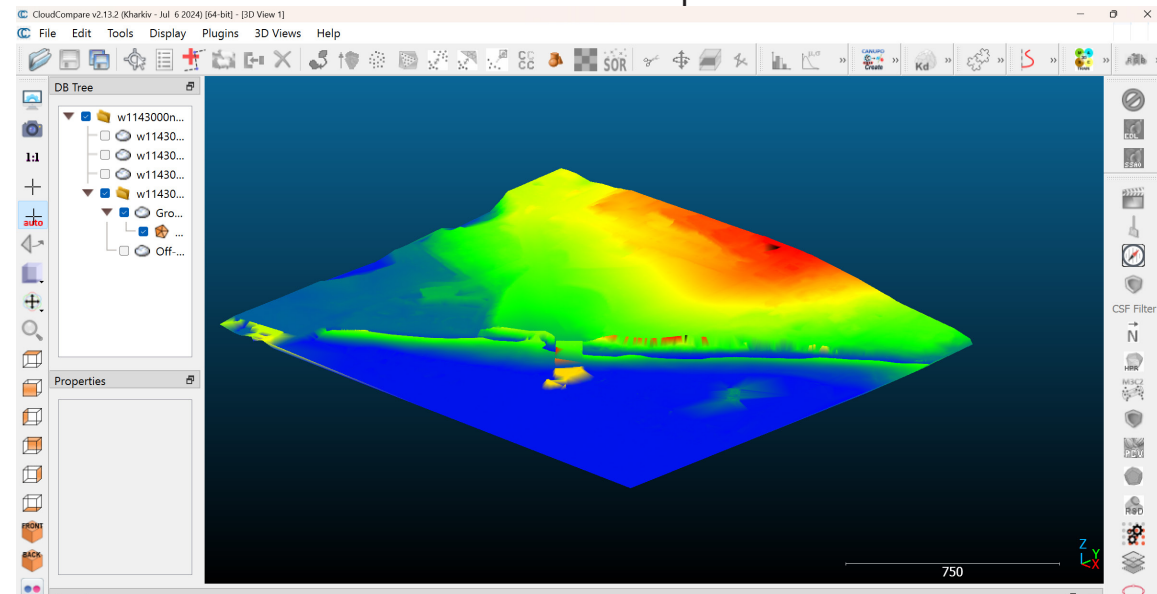
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Rhino Process:

5. Generate mesh in cloud compare

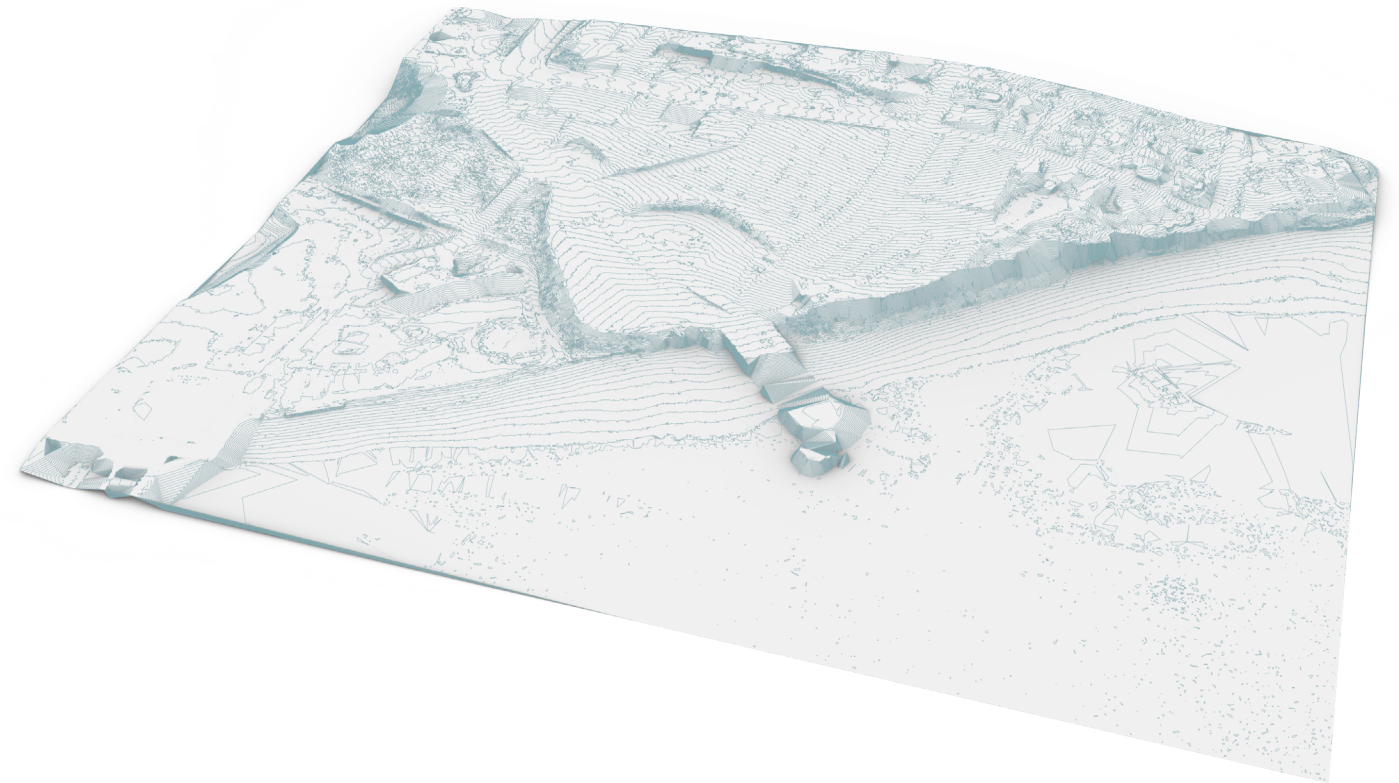


- > Select ground points
- > Edit
- > Mesh
- > Delaunay (xy Plane)

- > Save Laz Files
- > Export Mesh as OBJ

* The dock still looks kind of wonky but this is the best i could get it in CC.

6. Create Contours



Next steps:

- > Use grasshopper to clean up the noise from the really small contours
- > Organize script to create a laser cut file

* My file was crashing quite a bit because of the size of the file even after reducing the mesh. I plan to go back in and fix the dock and then continue with the grasshopper script provided for class.